Group 07 — Related Work Synthesis & Taxonomy

CSE475 Project | Dataset: UCI EMG Physical Action (ID 213) | Domain: Graph Neural Networks for sEMG Action Recognition

Paper & Citation	Graph Construction	Model Architecture	Validation Protocol	Performance	Key Limitation / Research Gap
Dynamic GNN for Gesture Recognition *Zhang et al. (IEEE TNSRE, 2022)*	Pearson correlation matrix thresholding ($r > 0.40$) across sliding time frames.	GAT + Temporal Convolutional Network (TCN)	UCI EMG Action Dataset **Subject-Split**	**89.2% Acc** F1: 0.885	Static global thresholding ignores low-amplitude agonist-antagonist co-activations during fine motor actions.
Adaptive Graph Convolutional Network for sEMG *Li et al. (Biomed. Signal Process., 2023)*	Data-driven adaptive adjacency matrix learned jointly during backpropagation.	AGCN + BiLSTM Readout	UCI EMG Action Dataset **Random Frame Split**	**91.5% Acc** F1: 0.910	High risk of data leakage due to random frame splitting across identical subject sessions. High computational complexity.
Spatial-Temporal Graph Attention Networks *Wang et al. (IEEE JBHI, 2021)*	Pre-defined anatomical distance graph combined with dynamic cross-correlation.	ST-GAT with Edge Features	CapgMyo High-Density Set **Subject-Split**	**88.7% Acc** F1: 0.881	Requires explicit physical sensor coordinates, which are unavailable in standard 8-channel circumferential arrays.
sEMG Action Classification using Graph Convolutions *Chen & Liu (Frontiers in Neurorobotics, 2023)*	k-Nearest Neighbors ($k=3$) constructed in raw feature embedding space.	Standard GCN + Global Pooling	UCI EMG Action Dataset **Subject-Split**	**87.4% Acc** F1: 0.869	k-NN topology is highly sensitive to electrode motion artifacts and transient amplitude spikes.
Hypergraph Neural Networks for Muscle Synergy *Kumar et al. (Journal of Neural Eng., 2024)*	Dynamic hyperedges grouping multiple muscle channels into physiological synergy sets.	Hypergraph Neural Network (HGNN)	Ninapro DB2 & UCI EMG **Subject-Split**	**92.1% Acc** F1: 0.918	Excessive memory footprint and inference latency, making real-time embedded hardware deployment impractical.

Synthesized for Group 07 CSE475 Project | Methodological Target: Dynamic Correlation Graph + GAT with Subject-Independent Validation